

Detection 1: Detection Theory & Single Measurement Detection

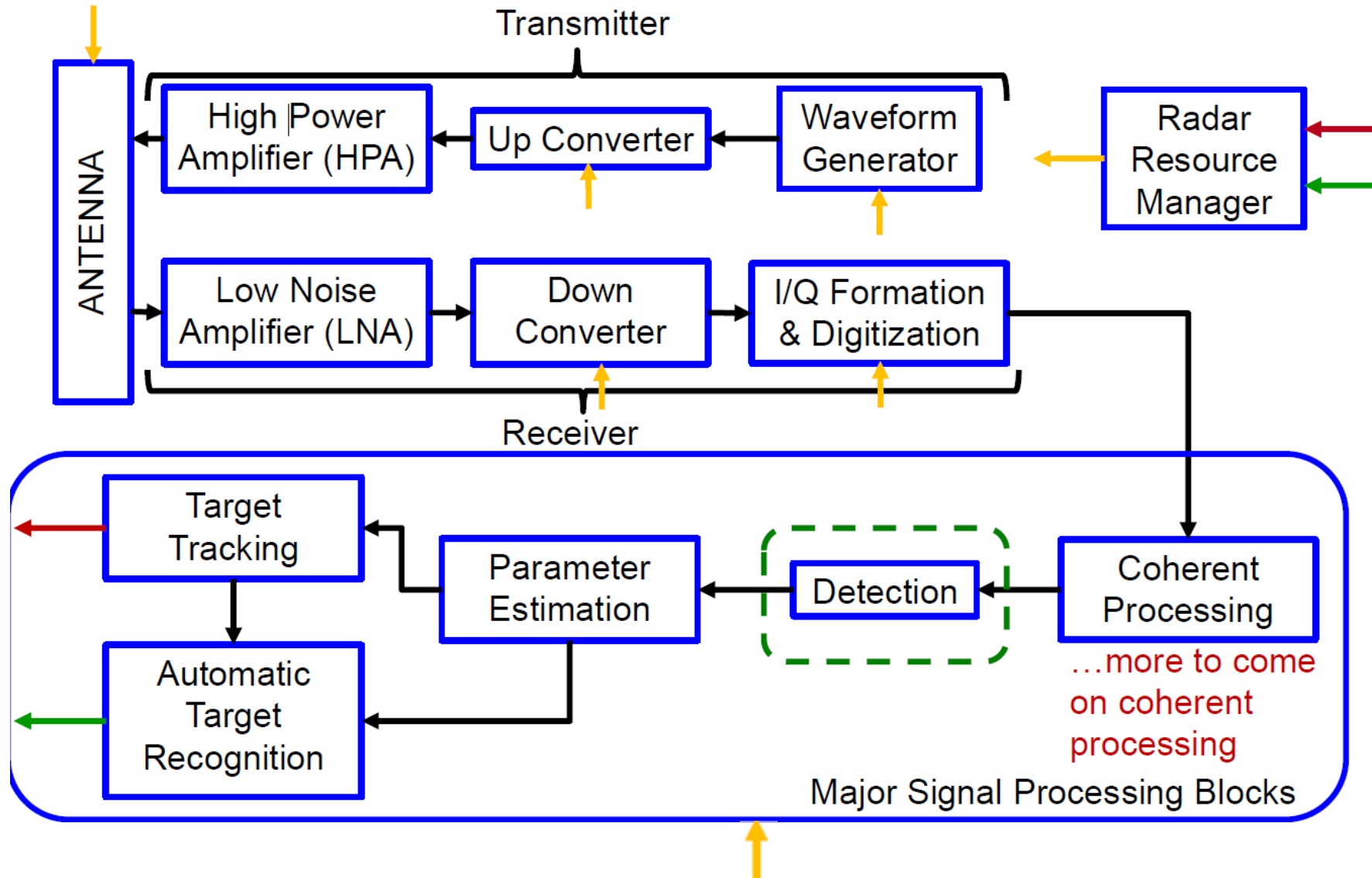
EE417 / EE517

CUA Spring 2024

Tuesdays 7:00-9:30pm

Instructor: Prof. Kevin Russo

Generic Radar System Block Diagram



“Single Measurement Detection” Means...

- We are only using $N=1$ Measurement in the threshold test
 - i.e. no coherent integration
- So we don't have to worry about the decorrelation properties of multiple measurements

Hypothesis Testing and the Likelihood Ratio Test

Testing the Data

- The fundamental jobs of a radar are **detection**, imaging, and tracking
- Data is first processed to maximize signal-to-interference ratio (SIR) – often coherent processing applied
- We then test the measured and processed data to decide if it
 - Represents interference only (“**hypothesis H_0** ”), or
 - Represents target+interference (“**hypothesis H_1** ”)
- If the latter, we will probably undertake more processing
 - *E.g.*, start a track on the detected target

Probabilities of Interest

- A given data sample represents a measurement of either interference only, or target + interference
- The data is a random process in either event
 - Even if the target is modeled as nonfluctuating, target+interference is still a random process
- Therefore, we evaluate the performance of our detection algorithm using statistical measures, mainly ...
- **Probability of Detection, P_D**
 - The probability that a target is declared (*i.e.*, we choose H_1) when a target is in fact present
- **Probability of False Alarm, P_{FA}**
 - The probability that a target is declared (*i.e.*, we choose H_1) when a target is in fact not present

Preview: Threshold Detection

- Almost all radars ultimately use **threshold detection** to decide whether a target is present
 - Results from solving an optimum “hypothesis testing” problem

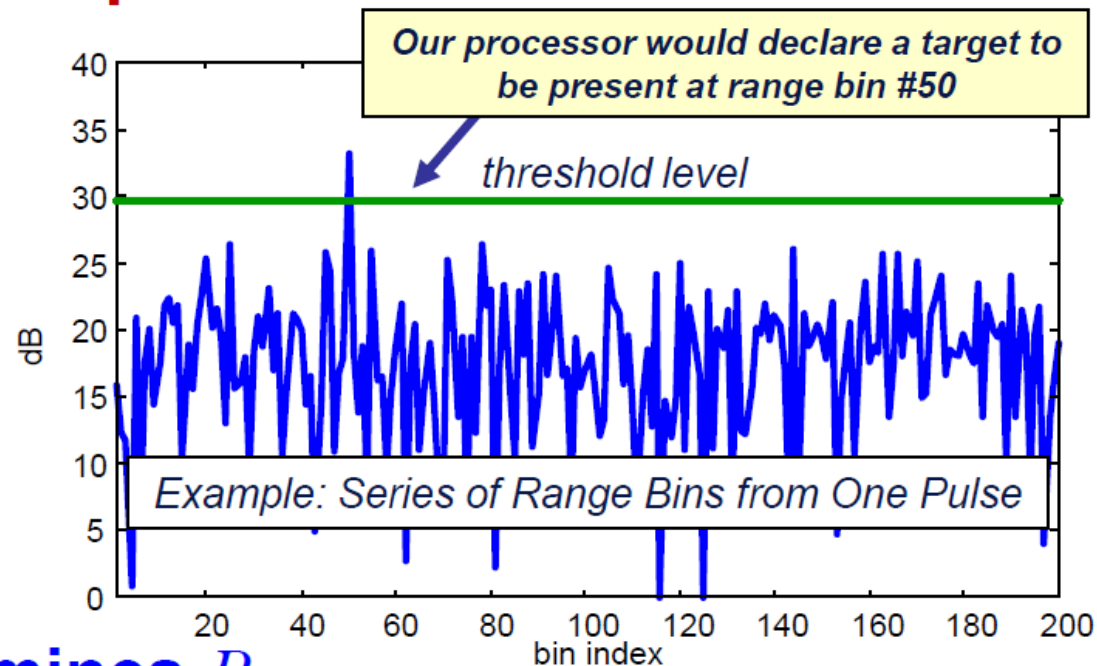
- **The threshold answers this question:**

- Is this data value “small” enough that it is probably just interference?

→ Choose hypothesis H_0

- Or “big” enough that it is probably **not** just interference?

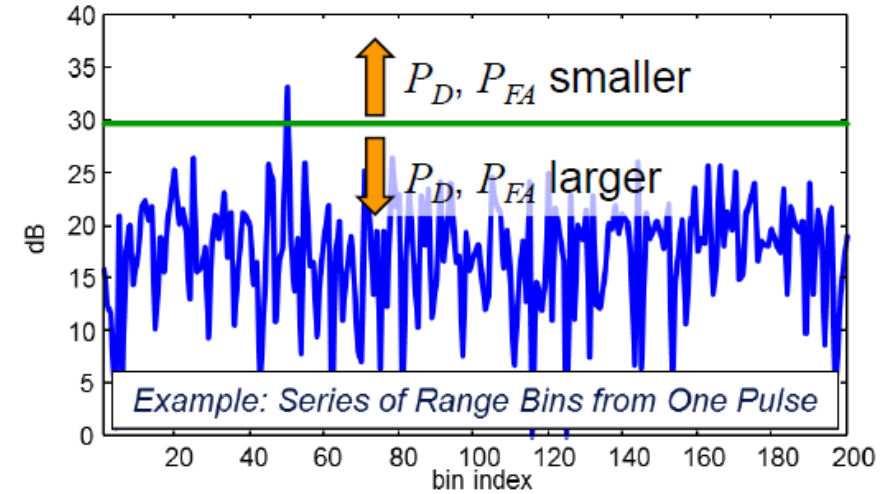
→ Choose hypothesis H_1



- The threshold level determines P_{FA}
- The threshold level + SIR determines P_D
 - SNR or SIR determines how far the target “sticks up” above the interference

Neyman-Pearson Criterion

- Higher threshold $\rightarrow$ lower P_{FA} (good), but also lower P_D (bad), and vice-versa
- So, given a measured data value, how do we strike the “best” balance between the two hypotheses?
- In radar, detection decisions are commonly based on the Neyman-Pearson optimization criterion
- **The Neyman-Pearson (N-P) rule:**
 - Set a fixed maximum value for P_{FA}
 - Generally flows down from system-level requirements
 - Often on the order of 10^{-3} down to 10^{-8}
 - Then maximize P_D



Likelihood Ratio Test (LRT) - 1

- Hypothesis testing under the Neyman-Pearson criterion leads to the **likelihood ratio test**, which in turn leads to threshold detectors
 - See the Richards' text [1] for the derivation
- Let y be the data sample to be tested
- We then need two PDFs
 - $p_y(y|H_0)$ = the PDF of 1 sample of interference if, in fact, a target is not present
 - Depends only on the interference PDF; targets are not involved
 - $p_y(y|H_1)$ = the PDF of 1 sample of target + interference if, in fact, a target is present
 - Will depend on the target fluctuation model and interference model
 - The two PDFs are obtained by analysis and/or experience with measured data

Likelihood Ratio Test (LRT) - 2

- Suppose I have 1 measured data sample and models for the two PDFs
- Evaluate each PDF by plugging your specific measured data value for y into the PDF formula
 - This gives you one number for $p_y(y|H_0)$ and one for $p_y(y|H_1)$
- The LRT tells us to decide “target present”, *i.e.* we choose H_1 , if

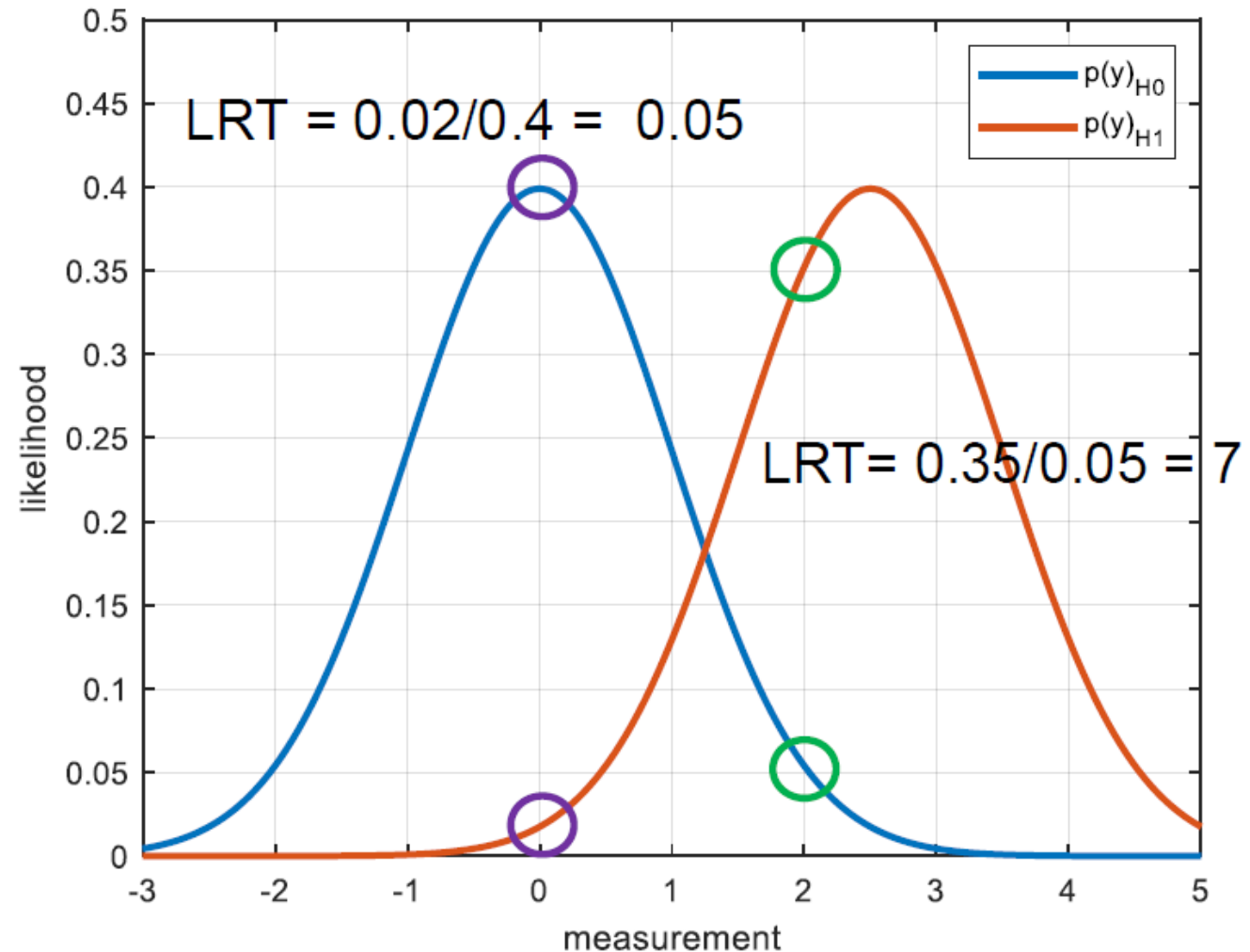
$$\frac{p_y(y|H_1)}{p_y(y|H_0)} \equiv \Lambda(y) \underset{H_0}{\overset{H_1}{>}} T_\lambda$$

(T_λ is a not-yet-specified threshold parameter)

- Otherwise, we declare target absent (choose H_0)
- Notice that this is a threshold test!

Visualizing the LRT

$$\frac{p_y(y|H_1)}{p_y(y|H_0)} \equiv \Lambda(y) \begin{matrix} > \\ < \end{matrix} \begin{matrix} H_1 \\ H_0 \end{matrix} T_\lambda$$



Log LRT

$$\frac{p_y(y | H_1)}{p_y(y | H_0)} \equiv \Lambda(y) \begin{matrix} > \\ < \end{matrix} \begin{matrix} H_1 \\ H_0 \end{matrix} T_\lambda$$

- The decision depends only on which number is larger
- We can perform any monotone increasing transformation on both sides of the LRT without affecting performance (combinations of P_{FA} and P_D)
 - e.g., if $\Lambda(y) > T_\lambda$, then $\sqrt{\Lambda(y)} > \sqrt{T_\lambda}$
- Most common is to use the log LRT, obtained by taking the natural logarithm of both sides:

$$\ln \Lambda(y) \begin{matrix} > \\ < \end{matrix} \begin{matrix} H_1 \\ H_0 \end{matrix} \ln T_\lambda$$

- The log LRT is frequently more algebraically convenient

Example: Nonfluctuating Target in Gaussian Noise

Example: Constant in Gaussian Noise

- **WARNING: Not a realistic radar example**
 - But very useful for illustrating procedures
- Detect the presence or absence of a constant in Gaussian noise
- Let w be a zero mean Gaussian noise sample with variance β^2
- **PDFs:**
 - $H_0: y = w \Rightarrow y \sim N(0, \beta^2)$ (Gaussian PDF with mean 0, variance β^2)
 - $H_1: y = m + w \Rightarrow y \sim N(m, \beta^2)$
 - $m > 0$
- m represents the signal voltage, β^2 represents the noise power, so the **single-sample SNR is m^2/β^2**

PDFs for a Constant in Gaussian Noise

- **Noise only (zero mean Gaussian noise):**

$$p_y(y|H_0) = \frac{1}{\sqrt{2\pi\beta^2}} \exp\left\{-\frac{1}{2}\left(\frac{y}{\beta}\right)^2\right\}$$

- **Target+noise (constant m + zero mean Gaussian noise):**

$$p_y(y|H_1) = \frac{1}{\sqrt{2\pi\beta^2}} \exp\left\{-\frac{1}{2}\left(\frac{y-m}{\beta}\right)^2\right\}$$

LR and Log LR for a Constant in Noise

- **The likelihood ratio is** $\Lambda(y) = p_y(y | H_1) / p_y(y | H_0) \rightarrow$

$$\Lambda(y) = \exp \left\{ -\frac{1}{2} \left(\frac{y-m}{\beta} \right)^2 \right\} / \exp \left\{ -\frac{1}{2} \left(\frac{y}{\beta} \right)^2 \right\}$$

- **Because of the form of the LR, the log LR is a natural here:**

$$\ln \Lambda(y) = -\frac{1}{2} \left(\frac{y-m}{\beta} \right)^2 + \frac{1}{2} \left(\frac{y}{\beta} \right)^2 = \frac{m}{\beta^2} y - \frac{m^2}{2\beta^2}$$

Log LRT

- Manipulate the LRT to get the data sample on one side of the equation, and all the constants on the other ...

$$\ln \Lambda(y) \underset{H_0}{\overset{H_1}{>}} \ln T_\lambda \Rightarrow$$

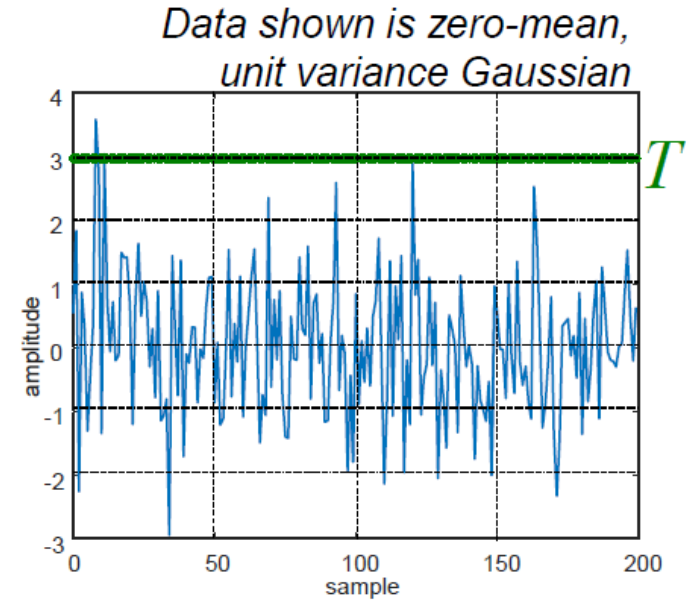
$$\frac{m}{\beta^2} y - \frac{m^2}{2\beta^2} \underset{H_0}{\overset{H_1}{>}} \ln T_\lambda \Rightarrow$$

$$y \underset{H_0}{\overset{H_1}{>}} \frac{\beta^2}{m} \ln T_\lambda + \frac{m}{2} \equiv T$$

- Roll all of those constants into a single new constant T

The Decision Rule for a Constant in Gaussian Noise

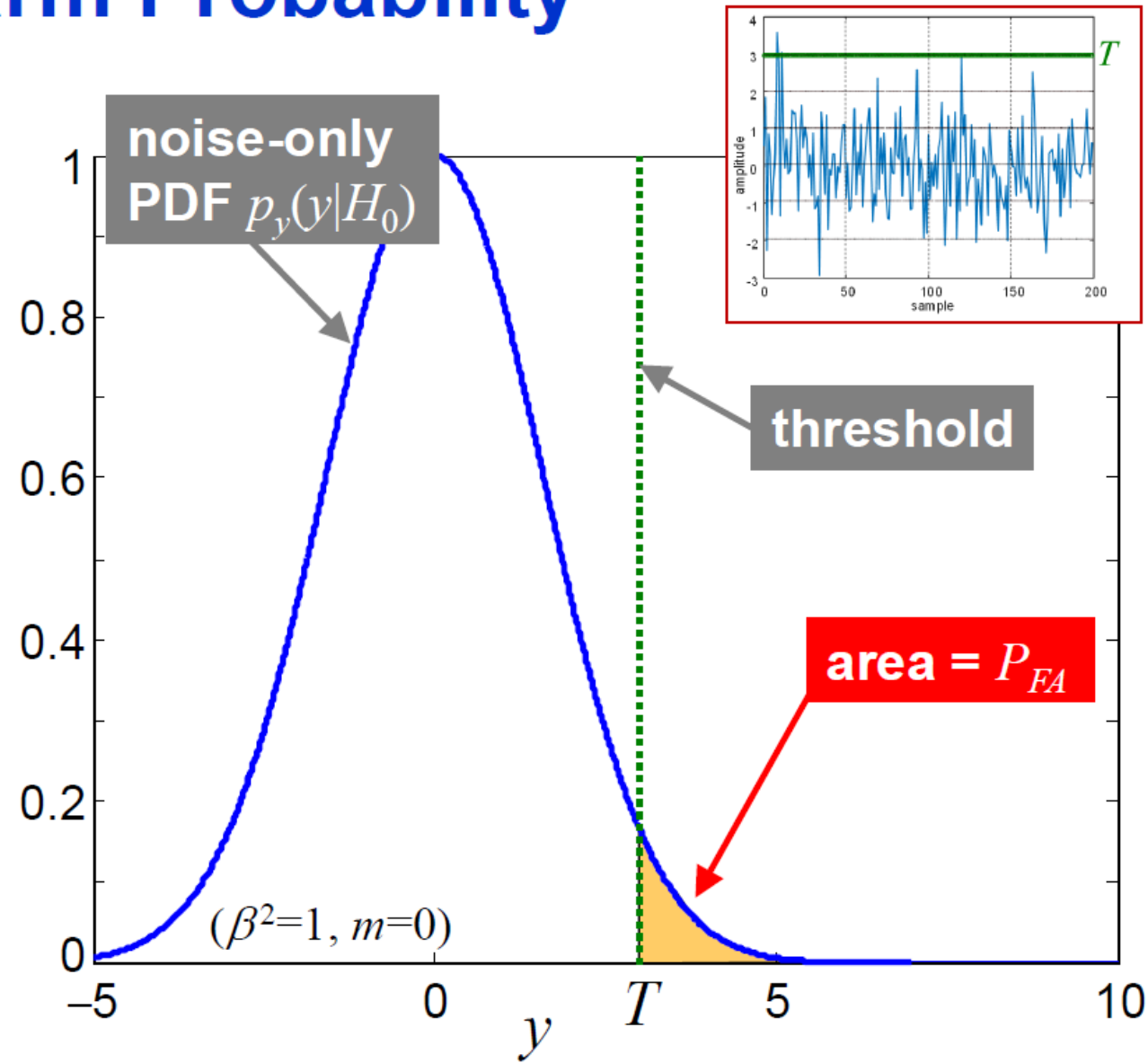
$$\begin{array}{c} H_1 \\ y > T \\ y < T \\ H_0 \end{array}$$



- All that LRT stuff reduces to the very reasonable rule “compare the data value to a threshold”!
 - Note that an LRT threshold test, when evaluated for specific PDF models, has turned back into a threshold test on the data itself
- The right-hand side is a constant
 - We don’t know its value yet
- The left hand side depends only on the measured data
 - Specifies the required data processing
 - In this case, no processing; just use the measured data value directly
- No need to specifically evaluate any PDFs in real time

False Alarm Probability

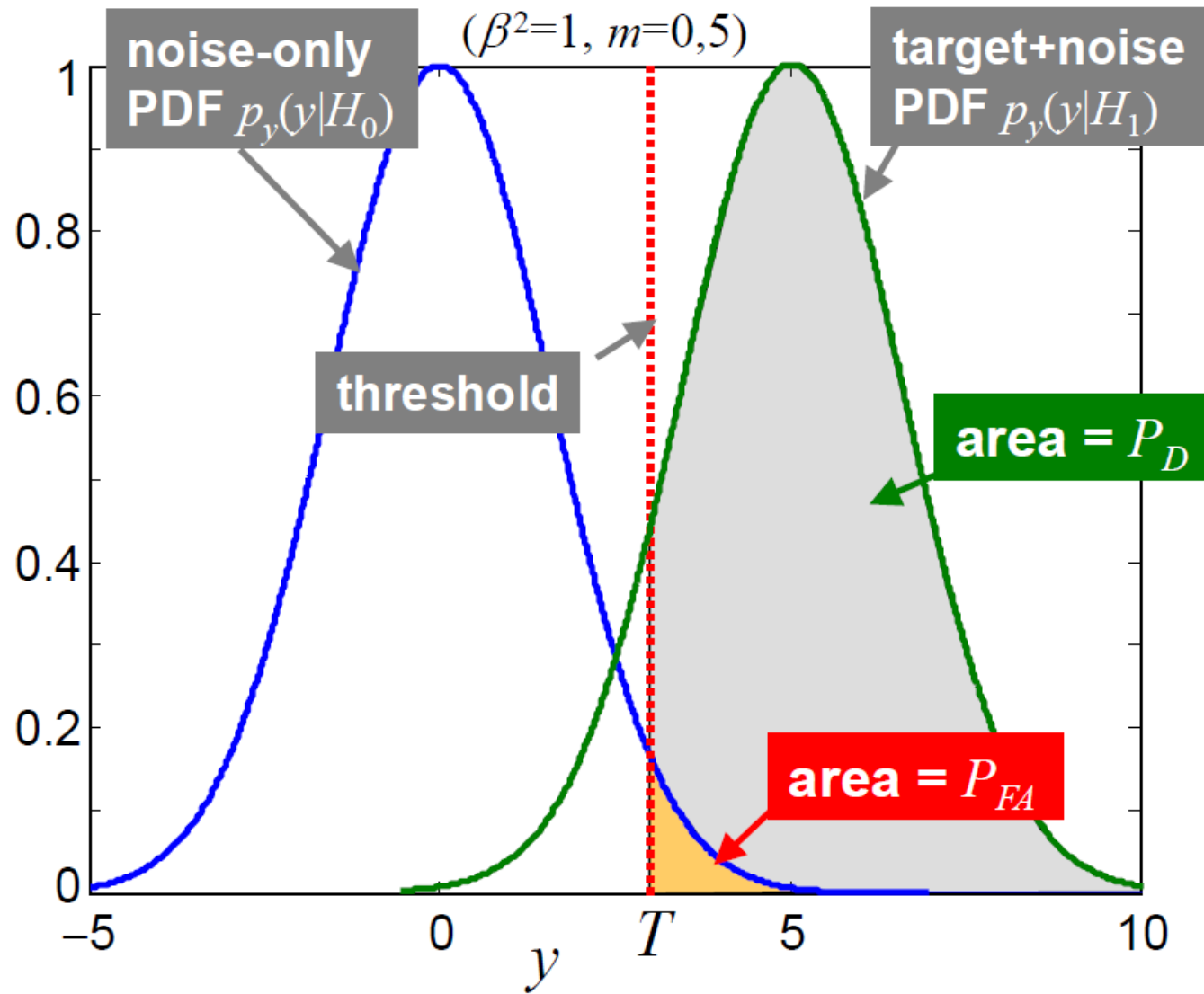
- P_{FA} is the probability that the signal crosses the threshold when there *really is not* a target, just interference
- The area under the $p_y(y|H_0)$ curve beyond the threshold gives P_{FA}
- Note that P_{FA} increases or decreases as the threshold decreases or increases



Note: this Gaussian PDF is not a good model for radar noise at the detector output

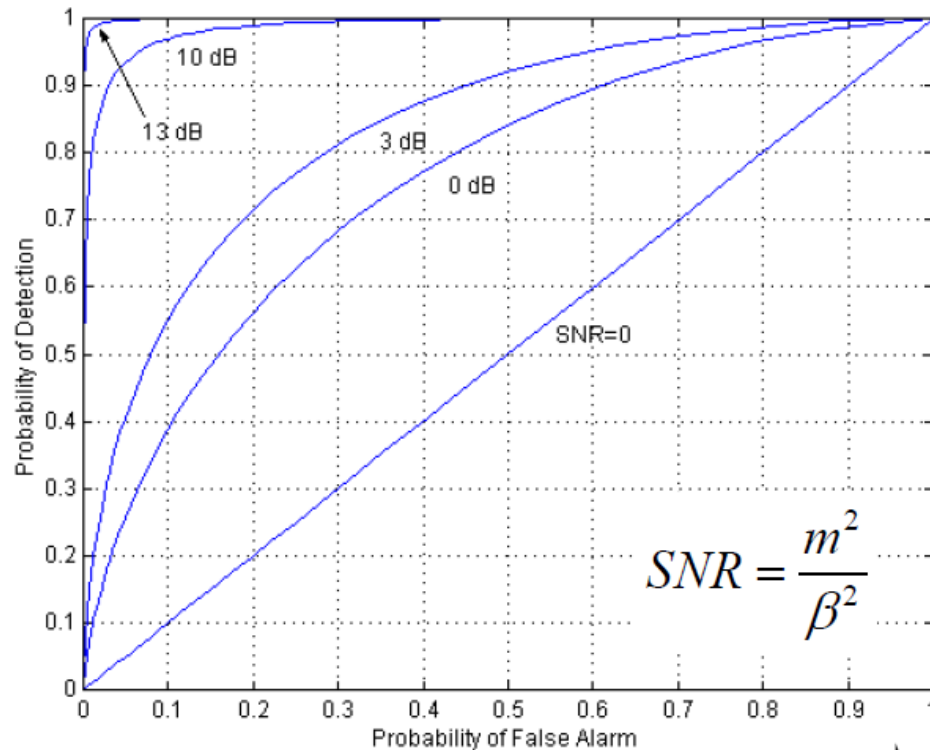
Detection and False Alarm Probabilities

- P_D is the probability that the signal crosses the threshold when there **really is** a target
 - We declare “target present”
- Note that P_D and P_{FA} increase or decrease together as the threshold decreases or increases
- The achievable combinations of P_D and P_{FA} are determined by the **overlap** of the PDFs



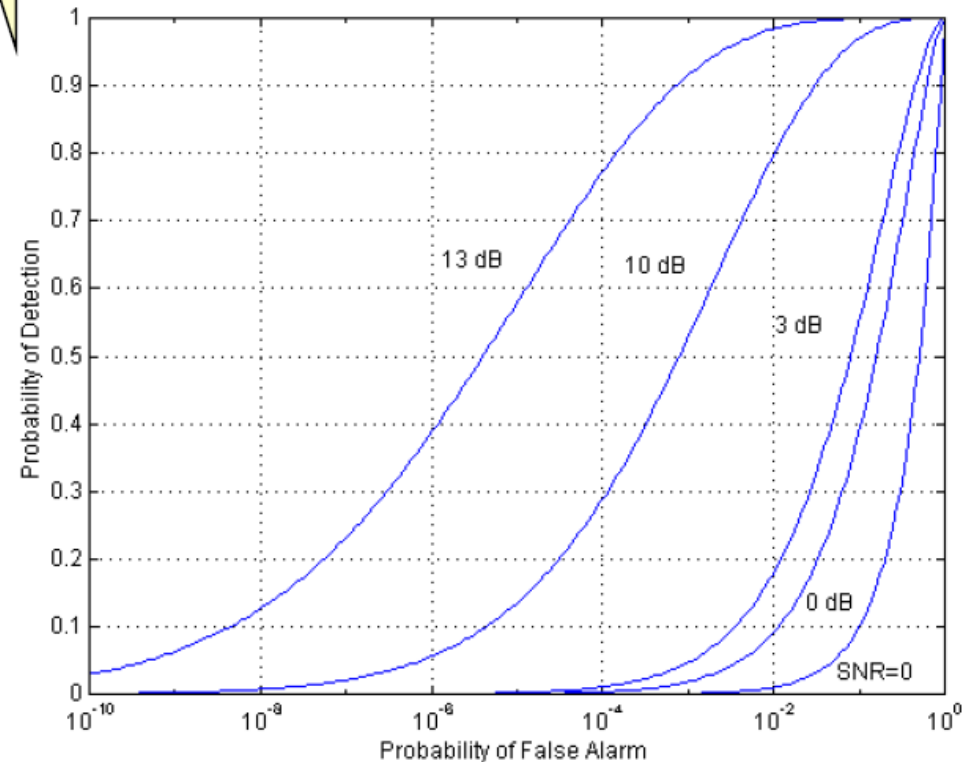
Note: these Gaussian PDFs are not a good model for radar data and are not Swerling models

Receiver Operating Characteristic



A log-log or semi-log scale is much better at showing performance for the low P_{FA} values of interest

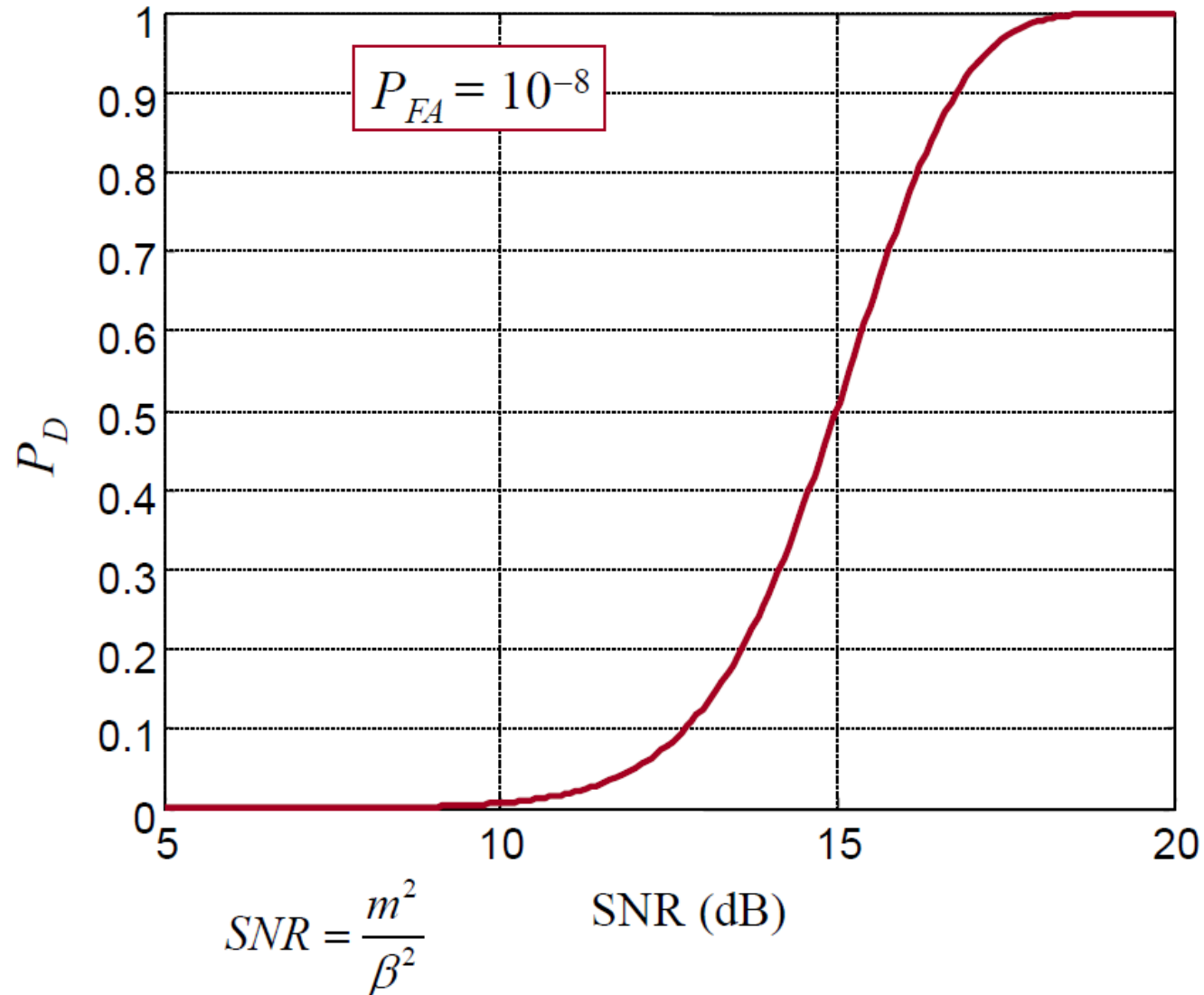
- ROC = “Receiver Operating Characteristic”
- Plot of two of P_D , P_{FA} , and SNR with the third as a parameter
 - This example is P_D vs. P_{FA} with SNR constant



– Same data in both plots; only difference is linear vs. log scale for the abscissa

Detection Performance for the Constant in Real Gaussian Noise Example

- Different form of the ROC from the previous page: P_D vs. SNR with P_{FA} held constant
- Notice that, for a given P_{FA} , P_D improves as the SNR goes up.
- Notice that P_D is
 - Bad for SNR < 10 dB,
 - Good for SNR > 18 dB
 - Gets decent for SNR in mid-teens of dB



How to Get Better Detection Statistics

- The goal of radar design for detection is to keep the “target present” and “target absent” PDFs as disjoint (nonoverlapping) as possible
 - This gives us a higher P_D for a given P_{FA}
- We can do this by modifying the PDFs in two ways:

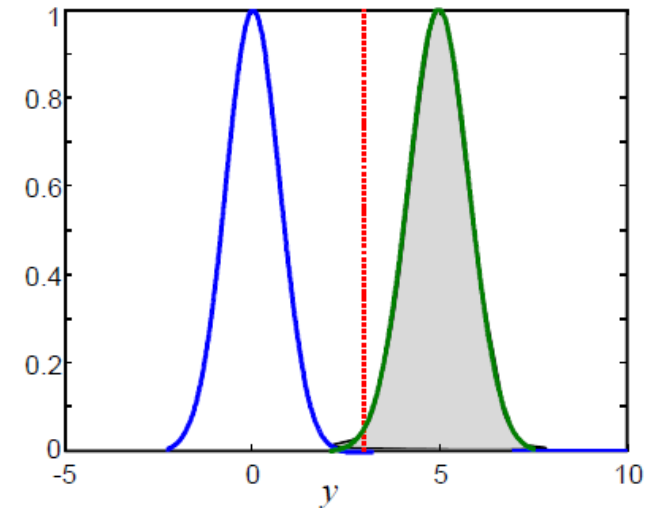
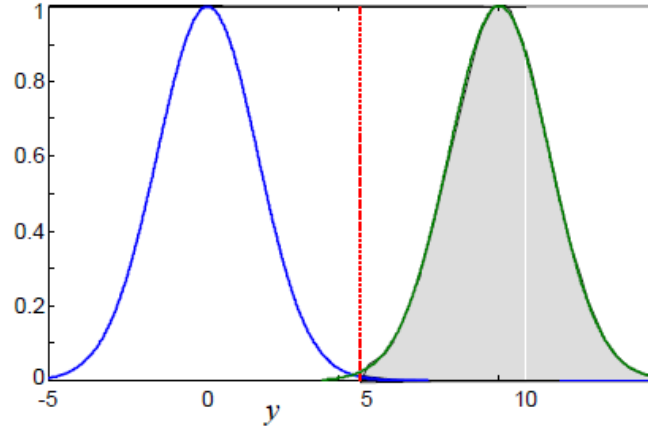
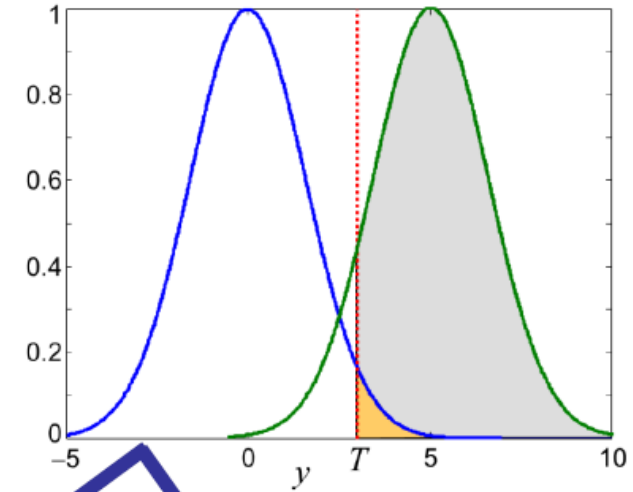
- Shift them further apart

- Requires a stronger target signal m

- Narrow their variance

- Requires a smaller noise power β^2

- **Either way is equivalent to increasing the SNR m^2/β^2**



$$SNR = \frac{m^2}{\beta^2}$$

Now for a More Realistic Example

Let's find the optimum receiver in the Neyman-Pearson sense for a

- **Single sample ($N = 1$) of a**
- **Nonfluctuating Target in**
- **Complex White Gaussian Noise**
using a
- **Coherent Receiver with a**
- **Square-Law Detector**

Repeat the Previous Process, But With ...

- A coherent receiver
 - Complex (I and Q) data
- Unknown **target** parameters
 - Amplitude, **absolute phase of echo**
- **Noise power (to set P_{FA}) (assumed known or estimated)**
- **Appropriate PDFs**

Absolute Phase

- A range change of $\lambda/4$ represents a two-way range change of $\lambda/2$ m, and therefore a phase change of 180°
- One wavelength is a small fraction of a range resolution cell
- It is usually unreasonable to expect that the absolute phase of a target echo can be known
 - That implies knowing the range to a small fraction of a range bin
- Therefore the target echo phase is assumed to be completely random
- So we need a receiver that doesn't depend on target echo phase information
- We get this by averaging the target+interference PDF over all possible target phase shifts
 - We can still coherently integrate samples related by structured (non-random) relative phase shifts due, e.g., to Doppler shift

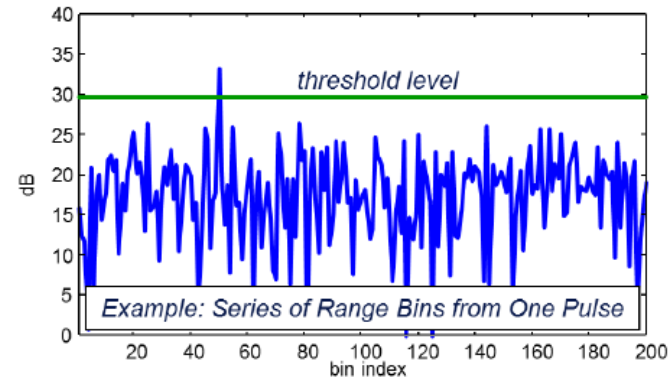
Optimum Detection Rule, Coherent Receiver and Nonfluctuating Target, $N = 1$

- The result is a **threshold test**:

$$\ln \left[I_0 \left(\frac{2|y|}{\beta^2} \right) \right] \begin{matrix} > \\ < \end{matrix} \begin{matrix} H_1 \\ T \\ H_0 \end{matrix}$$

I_0 = Modified Bessel function of first kind and order zero

$$I_0(z) = \frac{1}{2\pi} \int_0^\pi \exp(\pm z \cos \theta) d\theta$$



- The test depends only on the magnitude of y ; echo phase is not used
- But it requires some complicated transformations on the data before the threshold comparison

The Richards' text derives the detection rule

Alternate Detector Laws

- We don't want to compute $\ln\{y\}$ and $I_0\{y\}$ possibly millions of times per second
- We can substitute any monotone increasing function of y for $\ln\{I_0[y]\}$ without changing the outcome of the threshold test
 - E.g., if $y < T$, then $y^2 < T^2$ (assuming positive y)
- Most common are the:

Linear detector rule

$$\begin{array}{ccc} & H_1 & \\ |y| & > & T' \\ & H_0 & \end{array}$$

Square law detector rule

$$\begin{array}{ccc} & H_1 & \\ |y|^2 & > & T'' \\ & H_0 & \end{array}$$

Computing the Probability of False Alarm

- Coherent receiver, Gaussian noise, square-law detector, 1 measurement y
- The detection rule for this case is $|y|^2 \underset{H_0}{\overset{H_1}{>}} T$
- Let, $z = |y|^2$
- P_{FA} is the probability that the noise exceeds the threshold:

$$P_{FA} = \int_T^{\infty} p_{noise}(z) dz = \int_T^{\infty} p_z(z|H_0) dz$$

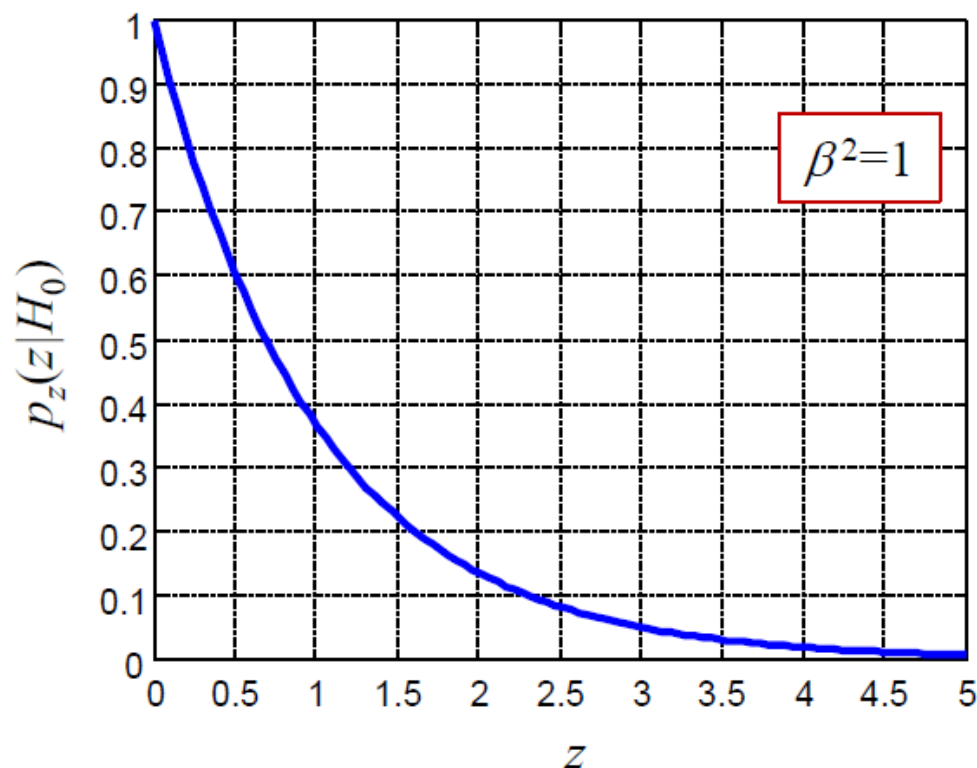
- But what is $p_z(z|H_0)$, the PDF of the squared-magnitude of a complex noise sample?

PDF of Detector Output for One Sample of Complex White Gaussian Noise

- Suppose the noise power is β^2
- Then the PDF of the squared-magnitude of a single complex noise sample z is

$$p_z(z|H_0) = \begin{cases} \frac{1}{\beta^2} e^{-z/\beta^2}, & z > 0 \\ 0, & z < 0 \end{cases}$$

- This is the **exponential PDF**



Probability of False Alarm Computation

- For a square-law detector and one data sample:

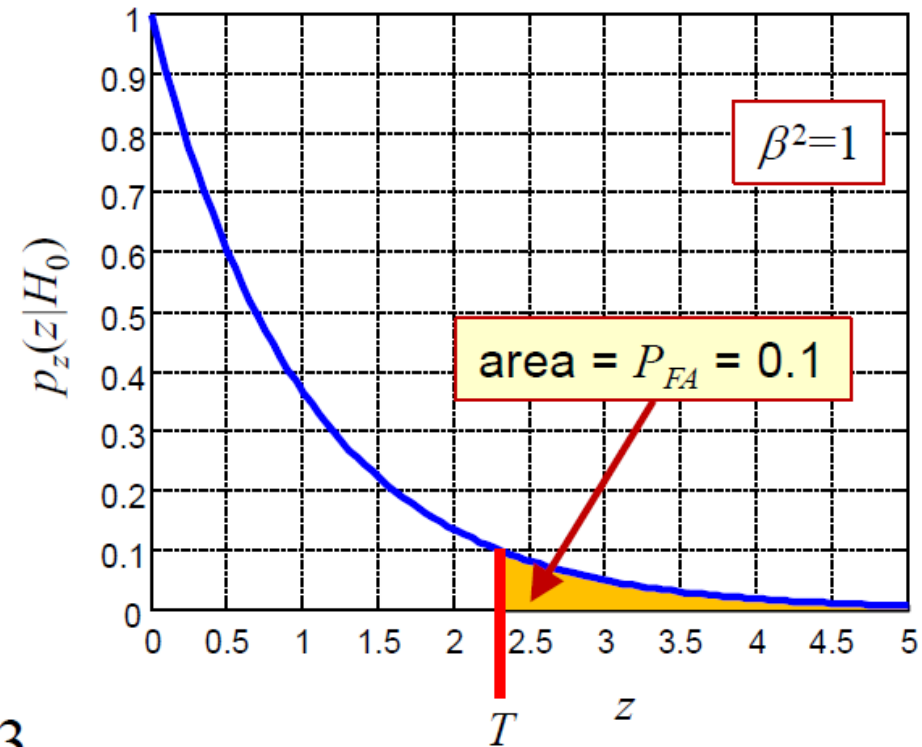
$$P_{FA} = \int_T^{\infty} \frac{1}{\beta^2} e^{-z/\beta^2} dz = e^{-T/\beta^2}$$

- Solving for T gives

$$P_{FA} = e^{-T/\beta^2} \Rightarrow T = -\beta^2 \ln(P_{FA})$$

- Note that the threshold is the noise power β^2 , multiplied by a factor that depends on P_{FA}

- Example: $\beta^2 = 1$, $P_{FA} = 0.1 \rightarrow T = 2.3$
 - Normal P_{FA} 's are much smaller $\rightarrow$ higher threshold



- OK, so now the threshold is set. What next?

Probability of Detection Calculation - 1

- Now we want to find out what P_D we get for that threshold
- The answer will depend on the SNR
 - So the threshold determines P_{FA} , then the SNR determines P_D
- To find P_D , we need the PDF of the data when a target is present, $p_z(z|H_1)$, so we can compute the threshold crossing probability
- That PDF is a noncentral chi-square of degree 2:

$$p_z(z|H_1) = e^{-z-\chi} I_0(2\sqrt{\chi z})$$

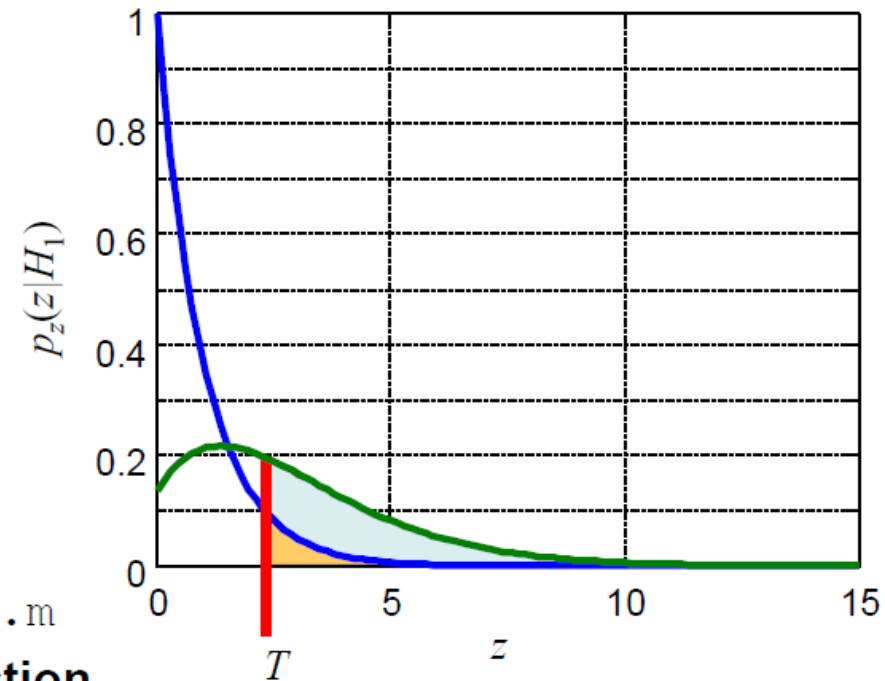
- Nonfluctuating target, white Gaussian noise, square law detector
- χ = SNR (linear scale, not dB!)
- I_0 is the modified Bessel function of first kind and order 0

Probability of Detection Calculation - 2

- Remember that the threshold T has already been set based on P_{FA} , so we can compute

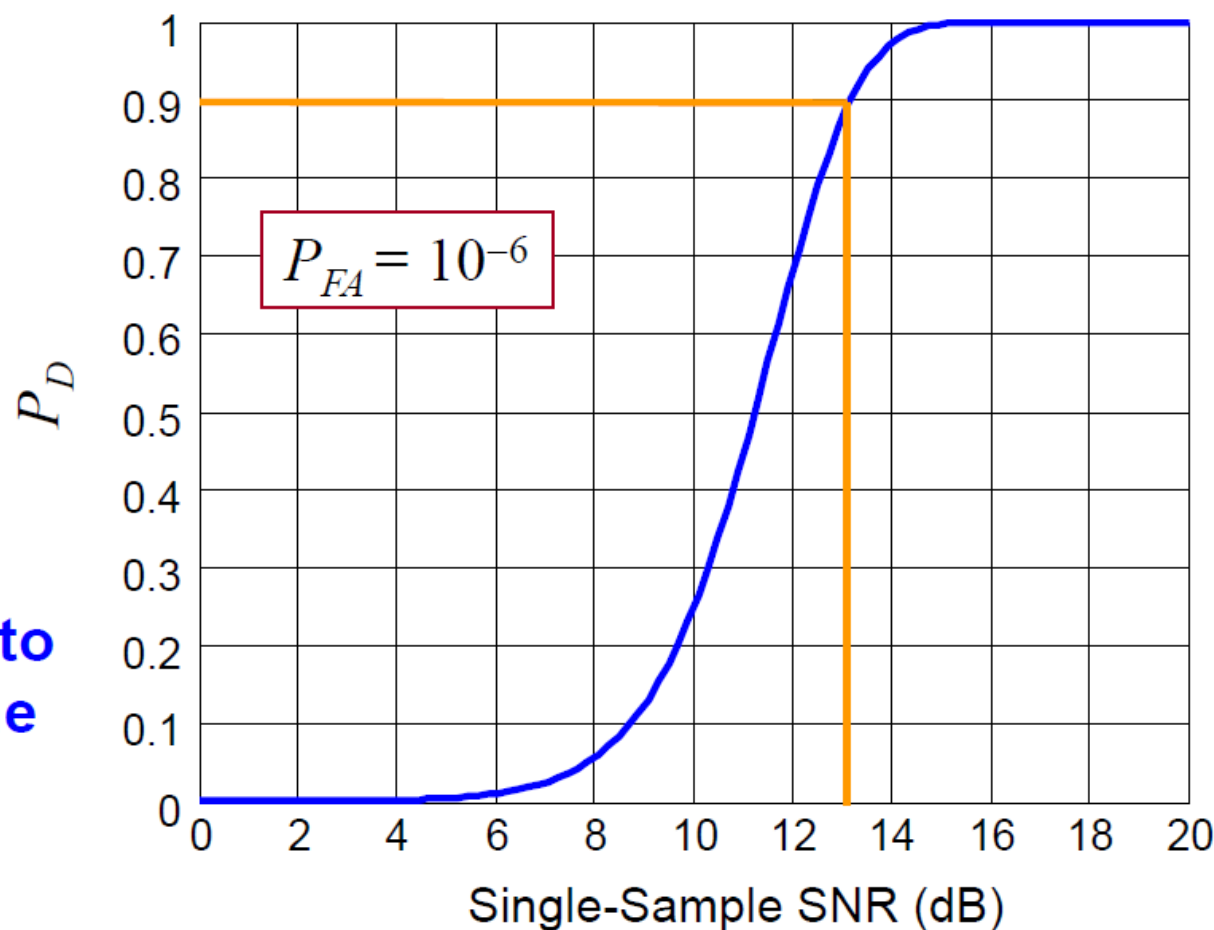
$$P_D = \int_T^{\infty} p_z(z|H_1) dz = \underbrace{Q(\sqrt{2\chi}, \sqrt{2T})}_{\text{Marcum's Q function}}$$

- The same example, continued, with SNR = 3 dB gives $P_D = 0.54$
 - Remember that normal thresholds are higher $\rightarrow$ we will need a higher SNR to get a good P_D
- MATLAB routines are available for the efficient computation of Marcum's Q function
 - MATLAB[®] Communications and Signal Processing Toolbox have `marcumq.m`
 - `marcum.m` in the Detection Calculator section of your M-files or at www.radarsp.com



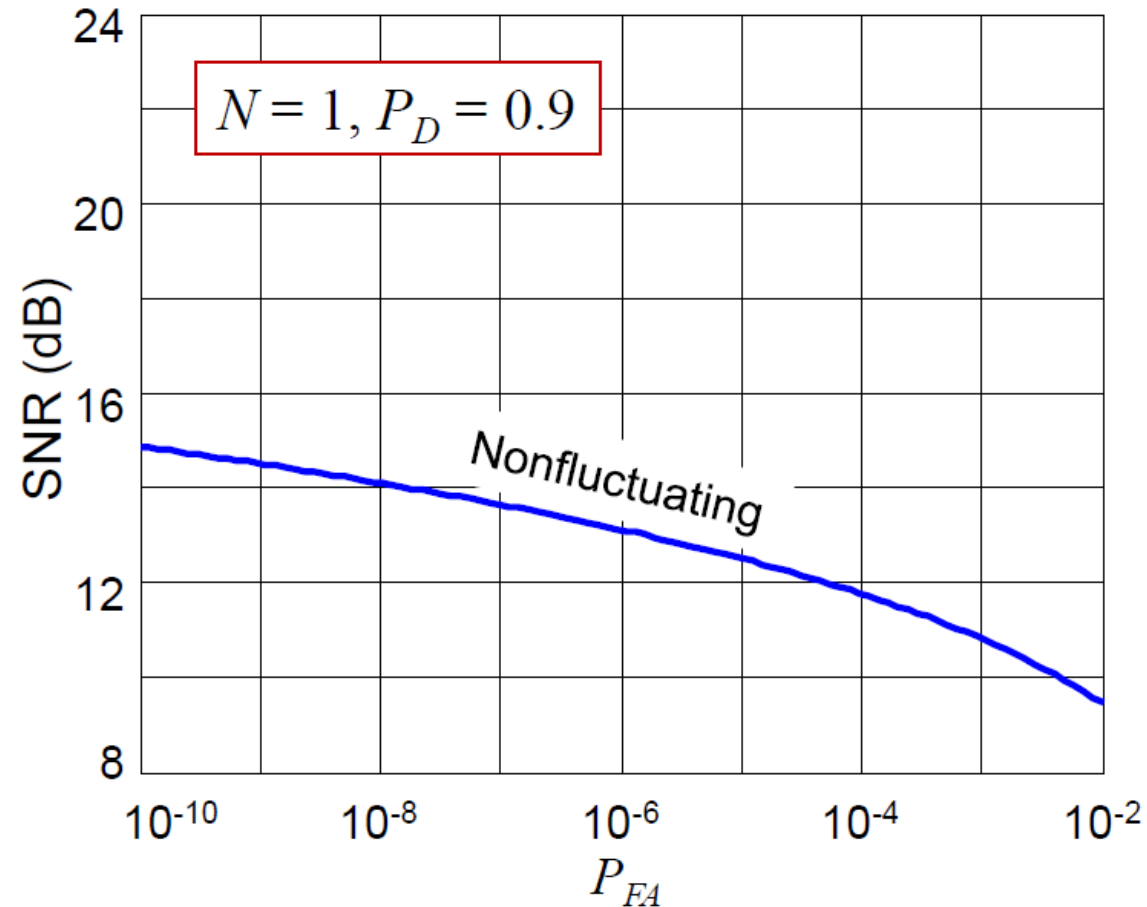
Detection Probability Increases with Increasing SNR

- Nonfluctuating target in complex Gaussian noise
- 1 sample
- $P_{FA} = 10^{-6}$
- Note that P_D is
 - Bad for SNR < 6 dB
 - Good for SNR > 14 dB
 - Changes rapidly in between
- Rule of thumb: SNR needs to be at least 13 dB or so at the “point of detection” for “decent” performance
 - Nonfluctuating target
 - Loose statement; use with caution!



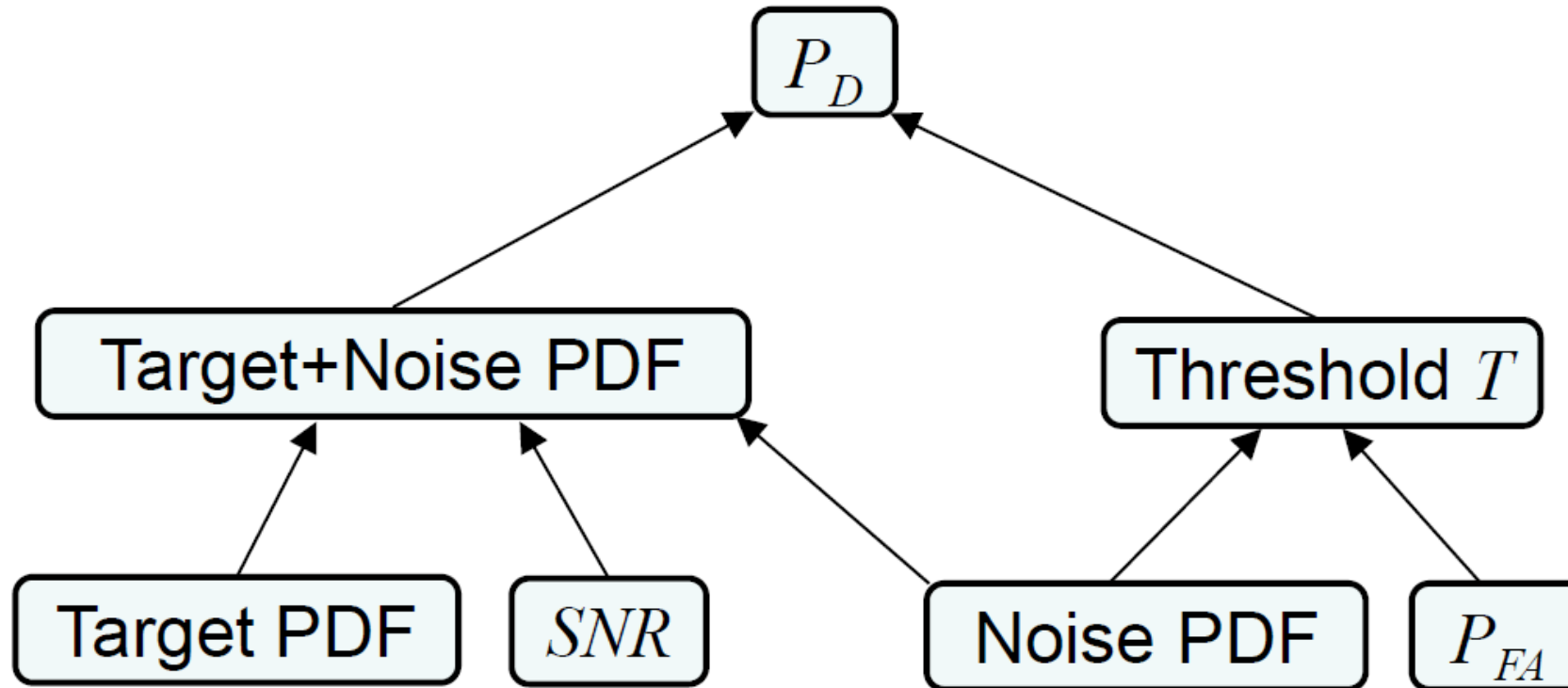
Variation of SNR Required for Good Detection Performance with P_{FA}

- **Nonfluctuating target**
- $P_D = 0.9$ on a single sample requires between about 11 and 15 dB, as P_{FA} ranges from about 10^{-3} to 10^{-10}
- **Around 13 dB is the middle of that range, still a good rough figure to know**
- **This curve computed using Shnidman's equation**
 - See the Appendix to the next block (“Detection 2”) of notes



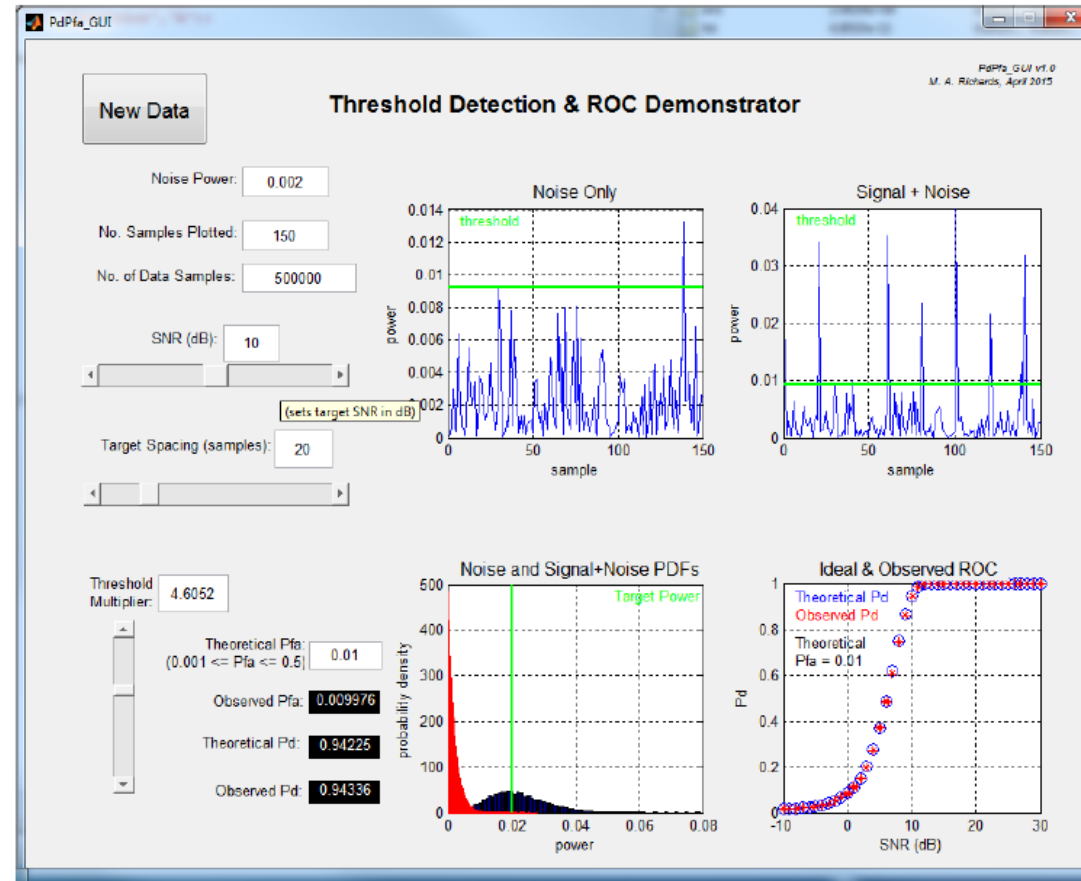
Relationship of Detection Variables

- Initial interference information: Desired P_{FA} , noise PDF at detector output
- Initial target information: target RCS PDF, SNR



PdPfa_GUI

- MATLAB GUI for detection using a single sample of a nonfluctuating target in white Gaussian noise
- Used to study the interactions of SNR, threshold, noise power, P_{FA} , and P_D
- To install and run:
 - Put PdPfa_GUI.m and PdPfa_GUI.fig in your MATLAB path
 - Type PdPfa_GUI at the MATLAB prompt



References

- M. A. Richards, *Fundamentals of Radar Signal Processing*, 2d. Ed. McGraw-Hill, New York 2014.
- M. A. Richards, J. A. Scheer, and W. A. Holm, eds., *Principles of Modern Radar: Basic Principles*. SciTech Publishing, 2010.
- Meyer, D. P., and H. A. Mayer, *Radar Target Detection*. (Academic Press, New York, 1973).
- DiFranco, J. V. and W. L. Rubin, *Radar Detection*. (Artech House, Dedham, Massachusetts, 1980.)
- Johnson, D. H. and D. E. Dudgeon, *Array Signal Processing*. (Prentice-Hall, Englewood Cliffs, New Jersey, 1993.)
- Kay, S. M., *Fundamentals of Statistical Signal Processing, Vol. II: Detection Theory*. (Prentice-Hall, Upper Saddle River, New Jersey, 1998.)
- Shnidman, D. A., "Radar Detection Probabilities and Their Calculation," *IEEE Trans. Aerospace and Electronic Systems*, vol. AES-31, no. 3, pp. 928-950, July 1995.
- Van Trees, H. L., *Detection, Estimation, and Modulation Theory, Part I: Detection, Estimation, and Linear Modulation Theory*. (Wiley, New York, 1968.)